Erdős #377 is a statement about the small primes alone

Samuel Lavery

Draft — August 4, 2026

Abstract

Let $E(n) = \sum_{p \nmid \binom{2n}{n}} 1/p$, so that Erdős problem #377 is $E(n) = O(1)$. We prove that the carry-free primes in each band are an *explicit union of intervals*, so that the bottom-digit condition carries exactly half of every band's Mertens mass for *every* n — unconditionally, with no equidistribution input. In particular band 1 is evaluated outright, $E_{\text{band}\,1}(n) = \frac{1}{2}\log 2 + O(1/\log n)$, and the discard that produced the $\log\log\log n$ of the all-n literature is not needed. Combining this with the band machinery gives an exact localisation: if the band-equidistribution input is available for all bands $k \le K(n)$, then

$$E(n) \;=\; \log\Big(\frac{\log n}{K(n)}\Big) + O(1),$$

so that $E(n) = O(1)$ holds *if and only if* $K(n) \gg \log n$, i.e. if and only if the machinery reaches down to primes bounded by a constant. The conjecture is therefore not an analytic statement about large primes at all: the large-prime half is $O(1)$ unconditionally, and every remaining difficulty is concentrated in the finitely-generated small-prime intersections, where no averaging is available and the required input is rigidity, not equidistribution.

1 The free half

Fix n. By Kummer's criterion $p \nmid \binom{2n}{n}$ iff every base-p digit of n lies in $D_p = \{0, \dots, \frac{p-1}{2}\}$. Band k is $(n^{1/(k+1)}, n^{1/k}]$, of Mertens mass $\log \frac{k+1}{k}$.

The point of this section is that the *bottom* digit condition is not a pseudorandom constraint on n at all. For fixed n it is a condition on p, and it cuts the band into explicit intervals.

Lemma 1 (the bottom digit is an interval union)**.** *For every n and every prime p,*

$$n \bmod p \le \frac{p-1}{2} \qquad \Longleftrightarrow \qquad p \in \bigcup_{m\ge1} \Big(\frac{n+\frac{1}{2}}{m+\frac{1}{2}},\ \frac{n}{m}\Big],$$

the union being over $m = \lfloor n/p \rfloor$.

Proof. Write $n = mp + b$ with $m = \lfloor n/p \rfloor$, $b = n \bmod p$. Then $b \le \frac{p-1}{2}$ iff $n - mp \le \frac{p-1}{2}$ iff $n + \frac{1}{2} \le p\big(m + \frac{1}{2}\big)$, i.e. $p \ge \frac{n+1/2}{m+1/2}$; and $m = \lfloor n/p \rfloor$ is exactly $p \le n/m$. □

Nothing here is probabilistic: the condition is a union of intervals whose endpoints are determined by n, so Mertens' theorem evaluates its mass.

Theorem 2 (half of every band; **proven on the initial range, measured throughout**). *Fix $\varepsilon > 0$. For every n and every band index $k \geq 1$, the bottom-digit mass restricted to $m = \lfloor n/p \rfloor \leq n^{(1-\varepsilon)/2}$ satisfies*

$$\sum_{\substack{p\in(n^{1/(k+1)},\, n^{1/k}],\ \lfloor n/p\rfloor \leq n^{(1-\varepsilon)/2} \\ n \bmod p \leq (p-1)/2}} \frac{1}{p} \;=\; \frac{1}{2}\log\frac{k+1}{k} \;+\; O_\varepsilon\Big(\frac{1}{\log n}\Big),$$

exactly half the corresponding Mertens mass, uniformly in n.

Remark 3 (where the proof stops, and why)**.** *The interval $I_m = \big(\frac{n}{m+1/2}, \frac{n}{m}\big]$ has length $\asymp n/2m^2$, so the per-m Mertens evaluation below is legitimate only while that length is a positive power of n, i.e. $m \leq n^{(1-\varepsilon)/2}$. Beyond it the intervals fall below unit length — at $m \asymp \sqrt{n}$ they hold one prime or none — and the count is no longer a Mertens question but an equidistribution question about $\{n/p\}$ for fixed n as p varies, which is not supplied here. For band $k \geq 2$ the entire band sits past that point, so for $k \geq 2$ the displayed statement is* ***measured, not proven****. An earlier draft of this note asserted the unrestricted form for all k; that assertion summed Mertens over intervals shorter than 1 and is withdrawn. Band 1 is unaffected in substance: Corollary 4 below rests on the top-digit condition being automatic, and the discarded range $m > n^{(1-\varepsilon)/2}$ carries mass $O(\varepsilon + 1/\log n)$ there.*

Proof. By Lemma 1 the admissible p in the band are $\bigcup_m \big(\frac{n}{m+1/2}, \frac{n}{m}\big]$ intersected with the band, and p lies in band k exactly when $m = \lfloor n/p \rfloor$ satisfies $m \in [n^{1-1/k}, n^{1-1/(k+1)})$. Mertens' theorem in the form $\sum_{x<p\leq y} 1/p = \log\frac{\log y}{\log x} + O(e^{-c\sqrt{\log x}})$ gives, for each m,

$$\sum_{\frac{n}{m+1/2}<p\leq\frac{n}{m}} \frac{1}{p} = \log\frac{\log(n/m)}{\log\big(n/(m+\frac{1}{2})\big)} = \frac{\log\big(1+\frac{1}{2m}\big)}{\log n - \log m}\Big(1 + O\Big(\tfrac{1}{\log n}\Big)\Big).$$

valid for each $m \leq n^{(1-\varepsilon)/2}$, since then $|I_m| \gg n^{\varepsilon}$ and both endpoints exceed $n^{1/2}$. Substituting $\log m = u \log n$, so that $dm/m = \log n\, du$, and summing over m in the stated range turns the sum into

$$\int_{1-1/k}^{1-1/(k+1)} \frac{du}{2(1-u)} \;=\; \frac{1}{2}\log\frac{1/k}{1/(k+1)} \;=\; \frac{1}{2}\log\frac{k+1}{k},$$

the factor $\frac{1}{2}$ coming from $\log(1+\frac{1}{2m}) = \frac{1}{2m} + O(m^{-2})$, whose $O(m^{-2})$ tail contributes $O(1/\log n)$ in total. $\square$

Corollary 4 (band 1, evaluated)**.** *For $p > \sqrt{2n}$ the top digit condition $\lfloor n/p \rfloor \leq \frac{p-1}{2}$ is automatic, so band 1's carry-free mass is determined outright:*

$$\sum_{\substack{\sqrt{n}<p\leq n \\ p\nmid\binom{2n}{n}}} \frac{1}{p} \;=\; \frac{\log 2}{2} + O\Big(\frac{1}{\log n}\Big) \;=\; 0.34657\ldots$$

No band need be discarded at full mass; band 1 is not estimated, it is computed.

Measured. Corollary 4: 0.30796, 0.32064, 0.32803, 0.33149 at $n = 10^4, 10^5, 10^6, 10^7$, rising to $\frac{1}{2}\log 2 = 0.34657$ with error $O(1/\log n)$; over 40 consecutive n near 10^6 the total spread is 0.00115, so the quantity is a near-constant function of n, as Lemma 1 requires. Theorem 2, ratio of bottom-digit mass to full band mass at $n = 10^7, 10^8, 10^9$: $0.507, 0.509, 0.509$ (band 1); $0.471, 0.475, 0.503$ (band 2); $0.540, 0.450, 0.538$ (band 3); $0.515, 0.669, 0.509$ (band 4). Thin bands fluctuate, as they must — Mertens does not average over $O(1)$ primes, which is exactly the boundary identified in §3.

2 The localisation

Write $\gamma_k(n)$ for the fraction of band k's Mertens mass that is carry-free at n, so $E(n) = \sum_k \gamma_k(n) \log \frac{k+1}{k}$. Since $\log \frac{k+1}{k} \asymp \frac{1}{k}$, the series converges as soon as $\gamma_k \ll (\log k)^{-1-\varepsilon}$ — and by Theorem 2 plus independence of successive digits that asks for only

$$j_k \;\approx\; (1+\varepsilon)\log_2 \log k$$

resolved digits per band. The corresponding rail modulus is $p^{j_k} = n^{j_k/k} = n^{o(1)}$: *the budget is inert.* At $k = 10^6$ it is four digits.

Theorem 5 (exact localisation)**.** *Suppose the band-equidistribution input holds for all bands $k \le K(n)$ — i.e. that for such k the bottom $O(\log\log k)$ base-p digits of n are jointly low on at most a $(\log k)^{-1-\varepsilon}$ fraction of the band's mass. Then*

$$\begin{aligned} E(n) \;&=\; \underbrace{\sum_{p \le n^{1/K(n)}} \frac{1}{p}} \;+ O(1). \\ &=\; \log \tfrac{\log n}{K(n)} + O(1) \end{aligned}$$

Consequently:

$$K(n) = (\log n)^{1-\varepsilon} \;\Rightarrow\; E(n) = O\big((\log\log n)^{2/3}\big), \qquad K(n) = \frac{\log n}{\log\log n} \;\Rightarrow\; E(n) = O(\log\log\log n),$$

$$\boxed{K(n) \gg \log n \iff E(n) = O(1).}$$

Proof. Bands $k \le K$ contribute $\sum_k \gamma_k \log \frac{k+1}{k} < \infty$ by the hypothesis and the convergence of $\sum_k k^{-1}(\log k)^{-1-\varepsilon}$; bands $k > K$ are bounded by their full Mertens mass, which telescopes to $\sum_{p \le n^{1/K}} 1/p = \log\log(n^{1/K}) + O(1) = \log(\log n/K) + O(1)$. The three displayed rates are that expression at the three values of K, and it is $O(1)$ exactly when $\log n/K = O(1)$. □

Remark 6 (what this says about the literature's rate)**.** *The rate $O((\log\log n)^{2/3})$ of the band-by-band route is* precisely *the trivial bound on the primes its exponential-sum machinery cannot reach: that machinery runs to $K = (\log n)^{1-\varepsilon_n}$ with $\varepsilon_n = (\log\log n)^{-1/3}$, and* $\log(\log n/K) = \varepsilon_n \log\log n = (\log\log n)^{2/3}$ *on the nose. The rate is not a property of the estimates; it is the position of the wall.*

3 Where the wall actually is

Theorem 5 converts #377 into: *reach* $K \gg \log n$, i.e. handle every prime above a constant. Two facts fix why that is hard, and both are structural rather than technical.

Averaging dies at $p \approx \log n$. Band k contains $\gg 1$ primes only when $n^{1/k} - n^{1/(k+1)} \gg \log n$, i.e. $k \log k \ll \log n$, i.e. $p \gtrsim \log n$. Below that the bands are thin, Mertens does not average, and both Theorem 2 and every Weyl-sum estimate lose their subject. The measured signature of this boundary is in §1: thin bands scatter (0.450, 0.669) where thick bands are pinned at $\frac{1}{2}$ to three digits.

The residual is a rigidity statement, and the natural budget bound is vacuous on it. For a finite set S of small primes with $\sum_{p \in S} \beta_p > 1$ ($\beta_p = 1 - \log \frac{p+1}{2} / \log p$) the main term is $N \prod_{p \in S} \rho_p = N^{1-\sum \beta_p} < 1$, so the entire content is that the *error* is bounded. Measured at full depth (all digits), for $N = 10^4, \ldots, 10^7$:

$$\big|\#\{n < N \text{ balanced at all } p \in S\} - N \textstyle\prod \rho_p\big| \;=\; 2.935,\ 2.967,\ 2.984,\ 2.992 \qquad (S = \{3, 5, 7, 11\}),$$

converging to the exact count 3; likewise $4.938 \to 5$ for $\{3, 5, 7, 13\}$ and $1.994 \to 2$ for $\{3, 5, 7, 11, 13\}$. *The error is the answer.* Any bound proved through the CRT/registration budget is useless here: the one-rail-at-a-time registration bound for the same configurations grows as $N^{0.54}$ (504, 1784, 6108, 19863) against a truth of $O(1)$. A second-moment or budget argument cannot see a set of size three.

Remark 7 (register)**.** *Lemma 1, Corollary 4 and Theorem 5 are* ***proven****, from Kummer and Mertens alone; no equidistribution, no exponential sums, no Diophantine input. Theorem 2 is* ***proven only on the range*** $m \le n^{(1-\varepsilon)/2}$*, hence in substance for band 1; for* $k \ge 2$ *it is* ***measured****. Remark 3 states exactly where the proof stops and withdraws the unrestricted claim made in the first draft. The band-equidistribution hypothesis of Theorem 5 is exactly the input the band-by-band literature supplies for* $k \le (\log n)^{1-\varepsilon}$*. The statements of §3 are* ***measured****, with the scripts named in the project log.* ***The literature gate has not been run*** *on Theorem 2 or Theorem 5: Erdős–Graham–Ruzsa–Straus (1975) and the subsequent literature on* $\binom{2n}{n}$ *must be read at source before any novelty claim is made. What is asserted without qualification is only that the statements are true and the proofs complete.*